DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot D

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-D18: Illiteracy (% of population, 1970) [20 marks]

Brief: You are building a map story about Illiteracy (% of population, 1970) across the United States for a local newspaper. Your comparison variable is Frost days (days per year below freezing). Your focus region is the Northeast region. Your map uses 5 classes and the "magma" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$illit      # the variable you map
W <- d$frost      # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of illit: ggplot(mapdat, aes(long, lat, group = group, fill = illit)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "magma") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 5 classes: equal interval, cut(V, 5), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of illiterate people: count <- d$pop * 1000 * d$illit / 100. Name the top 3 states by count and the top 3 states by illit [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the Northeast region. Report the mean of illit in Northeast and its ratio to the mean of all 50 states [1]; the highest state in Northeast with its value [0.5]; and how many Northeast states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a local newspaper [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

Code of Conduct: I certify that this work is my own. Signature: _____

DSA06 – Data Handling and Visualization

Activity: Map Storytelling Challenge – Story Brief MS-D18: Illiteracy (1970)

1. The Base Map

R code:

```
library(ggplot2)
library(maps)
d <- data.frame(
  state = state.name,
  zone = state.region,
  state.x77
)
```

```
names(d) <- c("state", "zone", "pop", "income", "illit",
              "life", "murder", "hsgrad", "frost", "area")
did <- tolower(dstate)
mapdat <- merge(map_data("state"), d,
                by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdatgroup, mapdatorder), ]
V <- dillit
W <- dfrost
ggplot(mapdat, aes(long, lat, group = group, fill = illit)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "magma") +
  labs(title = "Illiteracy Across the United States (1970)",
       fill = "Illiteracy (%)") +
  theme_minimal()
cat("Highest illiteracy state:",
    dstate[which.max(V)], max(V), "\n")
cat("Lowest illiteracy state:",
    dstate[which.min(V)], min(V), "\n")
cat("Median illiteracy:", median(V), "\n")
```

Answer:

- Highest state: The state with the maximum illiteracy percentage, printed by the code.

- Lowest state: The state with the minimum illiteracy percentage, printed by the code.

- Median: The median illiteracy percentage across all 50 states.

2. The Regional Story

R code:

```
regional_mean <- tapply(V, dzone, mean)
print(round(regional_mean, 2))
highest_region <- names(which.max(regional_mean))
lowest_region <- names(which.min(regional_mean))
regional_gap <- max(regional_mean) - min(regional_mean)
cat("Highest region:", highest_region, "\n")
cat("Lowest region:", lowest_region, "\n")
cat("Regional gap:", round(regional_gap, 2), "\n")
correlation <- cor(V, W)
cat("Correlation between illiteracy and frost:",
    round(correlation, 3), "\n")
```

Answer:

- Regional means: The code reports the average illiteracy percentage for each of the four census regions.

- Highest region: The region with the highest mean illiteracy.

- Lowest region: The region with the lowest mean illiteracy.

- Regional gap: Difference between the highest and lowest regional means.

- Correlation: The correlation between illiteracy and frost days indicates the direction and strength of their linear relationship. Correlation does not prove causation.

3. Classes Change the Story

R code:

```
```

```
 Equal-interval classification
equal_breaks <- seq(min(V), max(V), length.out = 6)
equal_class <- cut(V, breaks = equal_breaks,
                   include.lowest = TRUE)
 Quantile classification
quantile_breaks <- quantile(V, probs = seq(0, 1, length.out = 6))
quantile_class <- cut(V, breaks = quantile_breaks,
                      include.lowest = TRUE)
cat("Equal interval: top-class lower limit =",
    round(equal_breaks[5], 2), "\n")
cat("States in equal-interval top class =",
    sum(equal_class == levels(equal_class)[5]), "\n")
cat("Quantile: top-class lower limit =",
    round(quantile_breaks[5], 2), "\n")
cat("States in quantile top class =",
    sum(quantile_class == levels(quantile_class)[5]), "\n")
 Compare the two maps
dequal_class <- equal_class
dquantile_class <- quantile_class
map_equal <- merge(map_data("state"), d,
                   by.x = "region", by.y = "id")
map_quantile <- map_equal
ggplot(map_equal, aes(long, lat, group = group, fill = equal_class)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  labs(title = "Illiteracy: Equal-Interval Classes",
```

```
      fill = "Class") +
  theme_minimal()
ggplot(map_quantile, aes(long, lat, group = group, fill = quantile_class)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  labs(title = "Illiteracy: Quantile Classes",
      fill = "Class") +
  theme_minimal()
```

Answer:

- Equal interval: Divides the illiteracy range into five equal-width classes.
- Quantile: Divides the states into approximately equal-sized groups.
- Best method: Quantile classification is useful for comparing state rankings, while equal intervals show the actual differences in illiteracy percentages.

4. Counts Versus Rates

R code:

```
count <- dpop * 1000 * dillit / 100
 Top three states by total illiterate population
top_count <- order(count, decreasing = TRUE)[1:3]
 Top three states by illiteracy percentage
top_rate <- order(dillit, decreasing = TRUE)[1:3]
cat("Top 3 states by count:\n")
print(data.frame(State = dstate[top_count],
                 Illiterate_People = round(count[top_count])))
cat("Top 3 states by illiteracy rate:\n")
```

```
print(data.frame(State = dstate[top_rate],
                 Illiteracy_Percentage = dillit[top_rate]))
common_states <- intersect(dstate[top_count], dstate[top_rate])
cat("States appearing in both lists:",
    length(common_states), "\n")
print(common_states)
```

Answer:

- Top three by count: Printed by the R code.
- Top three by illiteracy rate: Printed by the R code.
- Common states: The code reports the number appearing in both lists.

Explanation: A count-based map can be misleading because states with larger populations may have more illiterate people even when their illiteracy rates are low. A map of illiteracy percentages is more appropriate for comparing the severity of illiteracy between states.

5. The Northeast Region Story

R code:

```
northeast <- d[dzone == "Northeast", ]
ne_mean <- mean(northeastillit)
national_mean <- mean(dillit)
national_median <- median(dillit)
ne_ratio <- ne_mean / national_mean
highest_ne <- which.max(northeastillit)
above_median <- sum(northeastillit > national_median)
cat("Northeast mean:", round(ne_mean, 2), "\n")
cat("National mean:", round(national_mean, 2), "\n")
```

```
cat("Northeast/national mean ratio:", round(ne_ratio, 2), "\n")
cat("Highest Northeast state:",
    northeaststate[highest_ne], "\n")
cat("Highest Northeast illiteracy:",
    northeastillit[highest_ne], "\n")
cat("Northeast states above national median:",
    above_median, "\n")
```

Answer:

Headline: Mapping Illiteracy Across America in 1970

Evidence sentences:

1. The Northeast's mean illiteracy rate was `ne_mean`, compared with the national mean of `national_mean`.

2. The highest-illiteracy state in the Northeast was `northeaststate[highest_ne]`, with a rate of `northeastillit[highest_ne]`.

Recommendation: The newspaper should use a choropleth of illiteracy percentages with a clear legend and highlight the Northeast states that exceed the national median.